2)

$(D+3)\,y_0(t) = 0 \qquad y_0'(0) = 2$

a) find and plot the exact solution for $0 \le t \le 10$

$r + 3 = 0$

$y_0(t) = C_1 e^{-3t}$

$y_0(0) = 2 = C_1$

$y(t) = 2e^{-3t}$

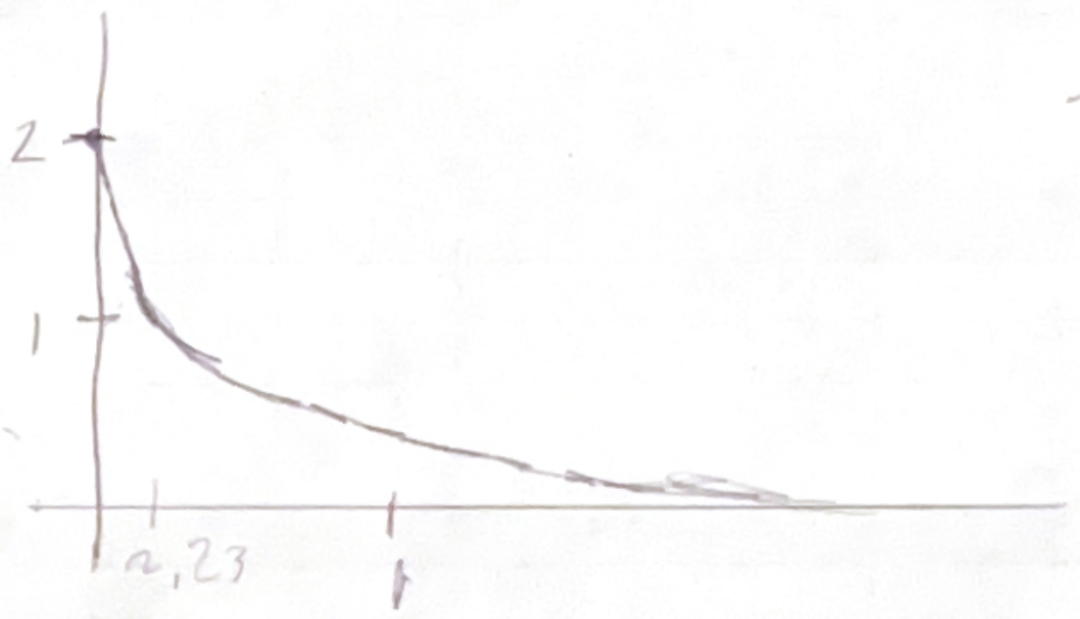
$2e^{-3t} = 1$

$e^{-3t} = \frac{1}{2}$

$-3t = \ln\frac{1}{2}$

$t = \frac{1}{3}\ln\frac{1}{2}$

$t = ,23$



c) compare results

using delta $t \cong ,01$ I was able to step through and get close to the graph above